

Assignment -1-

Design and Implementation of Linear Regression with Feature Rescaling and Gradient Descent for Optimal Prediction

Assignment Tasks

Task 1: Data Preparation

- Select or generate:
 - 1- One **univariate dataset** (e.g., house size vs price).
 - 2- One **multivariate dataset** (e.g., house size, number of rooms, location vs price).
- Perform:
 - 1- Data cleaning (handle missing values if any).
 - 2- Train-test split.

Task 2: Feature Rescaling

- Apply one or both:
 - 1- Normalization (Min-Max Scaling)
 - 2- Mean Normalization
 - 3- Standardization (Z-score)
- Compare results **with and without scaling**.

Task 3: Implement Linear Regression from Scratch

Do NOT use built-in regression models (e.g., from scikit-learn for training).

Task 4: Model Training

- Implement gradient descent:
 - Choose learning rate (α).
 - Experiment with different values of α .
 - Plot cost vs iterations.
- Train:
 - Univariate model
 - Multivariate model

Task 5: Model Evaluation

Evaluate your model using:

- Mean Squared Error (MSE)
- R^2 Score (optional but recommended)
- Compare:
 - With vs without feature scaling
 - Different learning rates

Task 6: Visualization

- Plot:
 - Regression line (for univariate case)
 - Cost function convergence
- For multivariate:
 - Use partial plots or predictions vs actual values

Task 7: Analysis and Discussion

Write a report discussing:

- Effect of feature scaling on convergence speed
- Impact of learning rate
- Differences between univariate and multivariate performance
- Challenges faced

Deliverables

You must submit:

1. **Source Code** (by using any language like Python, C++ etc..)
2. **Report (PDF or Word)** including:
 - Methodology
 - Results
 - Graphs
 - Analysis
3. **Dataset(s) used**

4. Output screenshots/plots

Notes

- Plagiarism is strictly prohibited.
- Code must be your own implementation.
- Proper documentation is required.